PROJECT DEVELOPMENT PHASE

Model Performance Test (Data Analytics)

Date:

19 June 2026

Team ID:

Project Name:

AI-Enhanced Intrusion Detection System

Maximum Marks:

10 Marks

Model Performance Testing

S.No.	Parameter	Screenshot / Values
1	Dashboard Design	Number of Visualizations / Graphs: **5** • Attack Distribution Chart • Normal vs Attack Traffic Chart • Prediction Statistics Dashboard • Confidence Score Visualization • Alert Activity Monitoring Chart Screenshot: Dashboard Page
2	Data Responsiveness	Dashboard updates prediction results and alert logs in real time. Average response time: **< 2 Seconds**. Screenshot: Real-Time Dashboard Output
3	Amount of Data Rendered (DB Metrics)	Dataset Used: NSL-KDD Dataset Records Processed: **10,000+ Network Traffic Samples** Features Processed: **10 Features** Screenshot: Dataset Statistics Page
4	Utilization of Data Filters	Filters Implemented: • Attack Type Filter • Confidence Score Filter • Date & Time Filter • Alert Severity Filter Screenshot: Alert Filtering Interface
5	Effective User Story	Number of Screens Added: **5** • Home

S.No.	Parameter	Screenshot / Values
		Page • Dashboard Page • Prediction Page • Alerts Page • About Page Screenshot: Application Navigation Screens
6	Descriptive Reports	Number of Visualizations / Graphs: **5** • Attack Trend Report • Intrusion Category Report • Prediction Confidence Analysis • Detection Summary Report • Security Monitoring Report Screenshot: Reports Dashboard

Dashboard Analytics Summary

Key Performance Indicators (KPIs)

KPI	Value
Total Records Processed	10,000+
Number of Features	10
Attack Categories Supported	5
Dashboard Pages	5
Visualizations	5
Average Response Time	< 2 Seconds
Model Accuracy	96.8%

Data Visualization Components

Dashboard Charts

1. Attack Category Distribution
2. Normal vs Malicious Traffic Ratio
3. Alert Frequency Analysis
4. Prediction Confidence Distribution
5. Real-Time Detection Statistics

Reports Generated

- Intrusion Detection Summary Report
- Attack Classification Report
- Prediction Accuracy Report
- Security Alert Report

- Monitoring Analytics Report

Conclusion

The AI-Enhanced Intrusion Detection System provides an effective data analytics dashboard that enables users to visualize intrusion detection results, monitor security alerts, and analyze network attack patterns. The dashboard offers multiple visualizations, real-time responsiveness, filtering capabilities, and descriptive reports, ensuring an efficient and user-friendly cybersecurity monitoring experience.